A Non-dominated Sorting Biased Random-Key Genetic Algorithm for the Multi-Objective Physical Cell Identity Assignment Problem

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Presentation Overview

1. Motivation & Context
2. Problem Definition
3. Proposed Method
4. Computational Experiments
5. Results & Analysis
6. Conclusion

Motivation
&
Context

Why PCI assignment is a hard (and practical) optimization task

Each Radio Access Network (RAN) cell must broadcast a **Physical Cell Identity (PCI)** so User Equipments (UEs) can detect and distinguish cells.

The PCI space is **limited** (e.g., **504** values in LTE; **1008** in 5G), so PCIs must be **reused**.

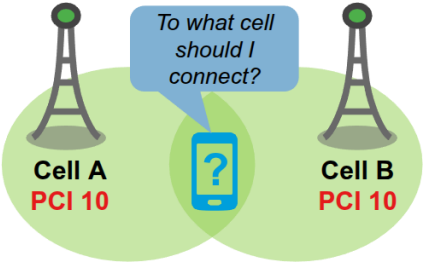
Poor reuse leads to technological conflicts (collisions, confusions, modulo-k interference) that degrade measurements and mobility.

Operators also care about **operational objectives**: minimize disruptive PCI changes (PCI shuffles) and avoid deployment failures during staged rollouts.

Technological constraints – Collisions

Neighboring cells must have different PCIs. Otherwise, a **collision** occurs, and the UE cannot connect to either cell. For example, UE in the figure cannot connect to a cell because both cells A and B have the same PCI 10.

Also, due to constraints in the primary and second sync signal and the 5G numerology, it is very desirable that neighboring PCIs be different in modules.

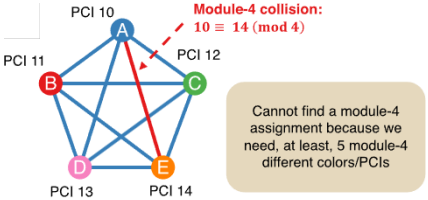


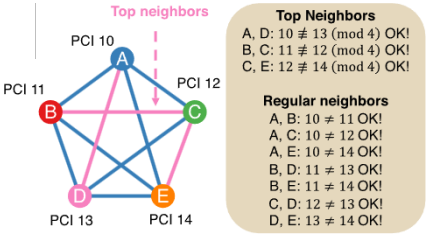
Technological constraints – Collisions

However, it is extremely hard to ensure module-3 and module-4 assignments in large production networks.

The reason is that it is very common to have a cluster of size ≥ 5 in which all cells are neighbors of the other cells with the cluster.

Precisely speaking, if we model these cells as nodes and the neighborhood relationships as edges in a neighboring graph, it is very common to find cliques of size $5+$.



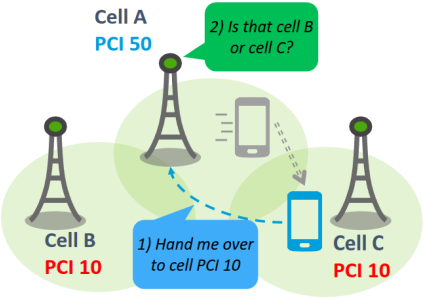


Technological constraints – Confusions

Confusion occurs when a cell has two neighbors with the same PCI.
When a UE asks for handover, the base station does not know to which cell itshould hand the UE over.

In other words, if **second-level neighbors (neighbor of neighbors)** have the same PCI that the current cell, confusion may occur.

Example: The UE is moving from cell A to cell C. Then, the UE asks Cell A to hand the connections over to the cell with PCI 10.
But now, Cell A cannot determine whether UE refers to Cell B or Cell C since they have the same PCI.



Operational constraints – Batch deployment

When scheduling batches, a PCI shuffle may occur: a node receives the PCI number previously assigned to a neighbor or a second-level neighbor, which is assigned to another batch.

Since the batches are deployed in sequence, we may find collisions and/or confusions, and therefore, the deployment will not be validated.

PCI shuffle – Example 1

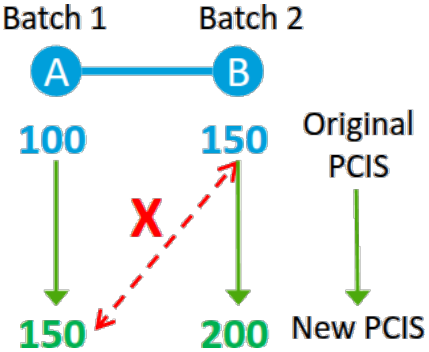
Suppose we have two neighboring cells A and B, which PCIs must be changed.

The original PCI for cell A is 100, and for cell B is 150.

Now, let us suppose that we must assign PCI 150 for cell A.

Since cells A and B are in different batches, we cannot deploy batch 1 first because it will cause a PCI collision with cell B, even though the PCI for B should be changed later on.

To avoid PCI shuffle, we must proper order the batches for delivery.



PCI shuffle – Example 2

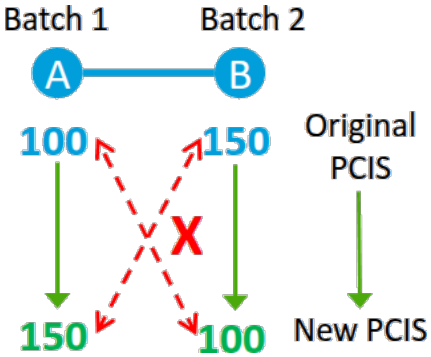
Here, PCI 150 must be assigned to cell A and PCI 100 must be assigned to cell B (crossed PCI shuffle).

In this case, it does not matter the other we try to deploy the batches because collisions will happen in any case.

To avoid PCI shuffle we must assign dependent nodes (for which we must change the PCIs) in the same batch.

However, each batch has a limited size or capacity.

Therefore, it can be challenging to find a proper packing.



Main Contributions

1. We propose a **Non-dominated Sorting BRKGA (NS-BRKGA)** to solve the **multi-objective** PCI assignment problem with conflicting objectives.
2. We design a **random-key encoding + specialized decoder** that combines:
   - fast best-effort **constraint repair**,
   - **lexicographically guided** local search, and
   - **genotype feedback** (re-encoding improvements into chromosomes).
3. We benchmark on **291 real-world instances** against five representative MOEAs (incl. NSGA-II), showing **stronger Pareto fronts** under the same time budget.

Problem Definition

Modeling the network and neighborhoods

We model the network as an undirected graph $G = (V, E)$:

- V: cells;
- E: neighbor/interference relations.

The edge set is partitioned:

$$E = T \cup N, \qquad T \cap N = \emptyset,$$

- T: **top-neighbors** (critical relations) where modulo-k collisions are *forbidden*;
- N: regular neighbors where direct collisions are forbidden, and modulo-k collisions are undesirable.

Second-level neighbors (used to model confusions):

$$S = \{(u, w) \mid (u, v) \in E,\ (v, w) \in E,\ (u, w) \notin E\}.$$

Decision Variables and Hard Constraints

Decision variable: assign each cell $v \in V$ an integer PCI $x_v \in AP \subseteq \{0, 1, \ldots, PCI_{\max}\}$.

Hard constraints (must hold):

1. **Collision-freeness** (no adjacent equal PCIs):

$$x_u \neq x_v, \quad \forall (u, v) \in E.$$

2. **Top-neighbor modulo-k collision-freeness**:

$$x_u \not\equiv x_v \pmod{k}, \quad \forall (u, v) \in T.$$

3. **Confusion-freeness** (no two second-level neighbors share a PCI):

$$x_u \neq x_v, \quad \forall (u, v) \in S.$$

Objective Functions

Let $\mathbb{I}(\cdot)$ be the indicator function and O_v the original PCI of cell v.

1. **Modulo-k collisions on regular neighbors** (interference):

$$f_1(x) = \sum_{(u,v)\in N} \mathbb{I}\big(x_u \equiv x_v \pmod{k}\big).$$

2. **PCI shuffles** (deployment safety during batched rollouts):

$$f_2(x) = \sum_{(u,v)\in E\cup S} \Big(\mathbb{I}(x_u = O_v) + \mathbb{I}(x_v = O_u)\Big).$$

3. **Total PCI changes** (stability / renumbering cost):

$$f_3(x) = \sum_{v\in V} \mathbb{I}(x_v \neq O_v).$$

Goal: approximate the **Pareto front** for (f_1, f_2, f_3) over feasible assignments.

Proposed Method: NS-BRKGA

https://github.com/luishpmendes/ns-brkga

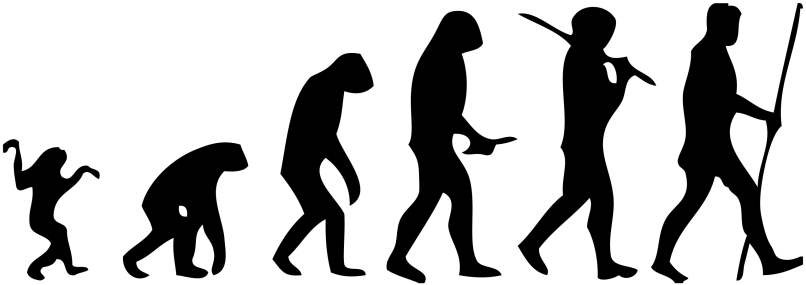
Non-dominated Sorting Biased Random-Key Genetic Algorithm

Genetic Algorithm (GA)

Paradigm that mimics natural evolution.

Analogy between a set of solutions and a population of individuals.

Each individual is coded as a chromosome (string of genes).



Non-dominated Sorting Biased Random-Key Genetic Algorithm

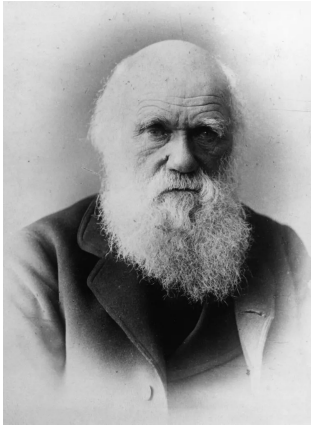
Genetic Algorithm (GA)

Each individual has a fitness value related to the solution quality.

The population evolves over several generations.

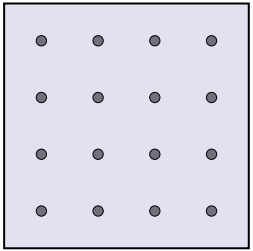
A new population is created using genetic operators (crossover and mutation).

“Survival of the fittest”

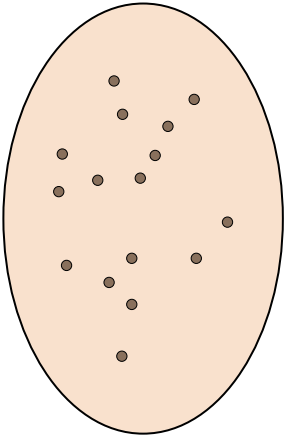


Non-dominated Sorting Biased Random-Key Genetic Algorithm

Random-Key Genetic Algorithm (RKGA)



$[0, 1)^n$



$\mathcal{X}$

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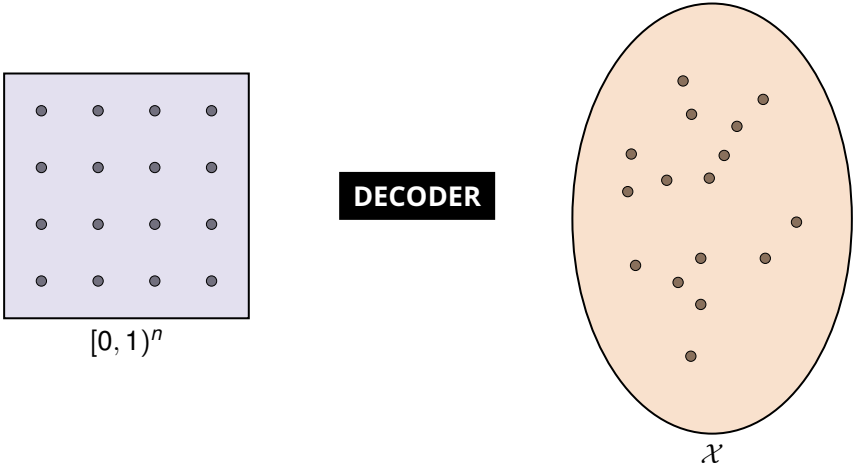

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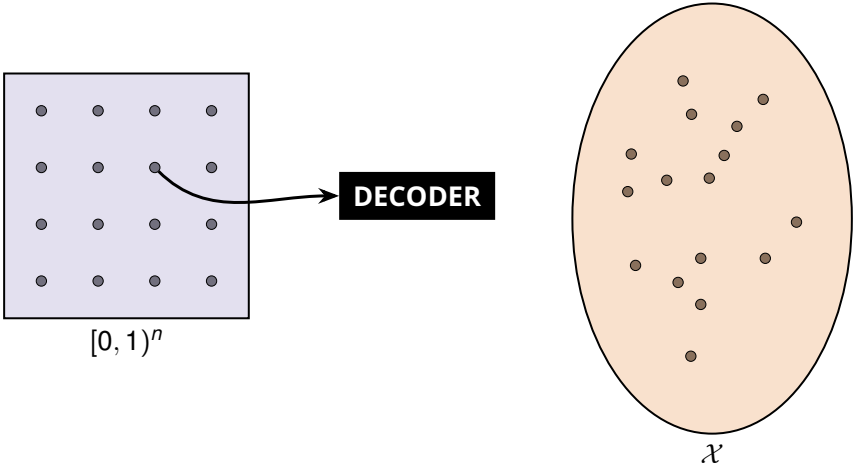
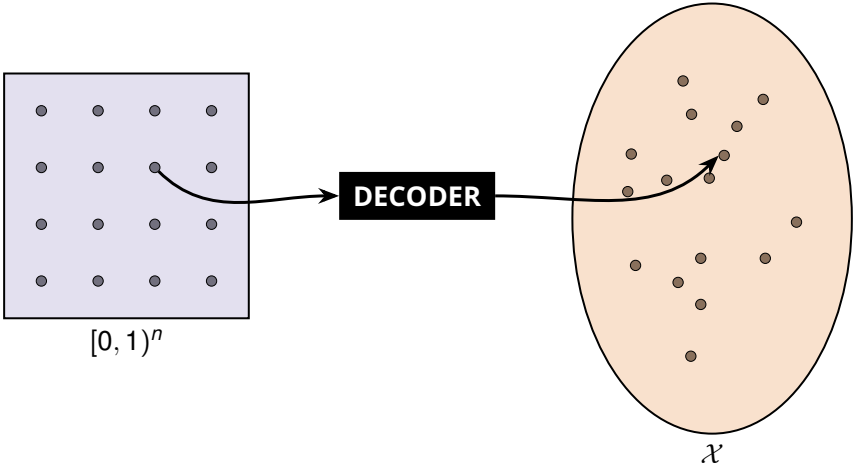


Non-dominated Sorting Biased Random-Key Genetic Algorithm

Random-Key Genetic Algorithm (RKGA)



Non-dominated Sorting Biased Random-Key Genetic Algorithm

Solution representation for MOPCI

A chromosome is a vector of random keys:

$$\mathbf{k} = (k_v)_{v \in V}, \qquad k_v \in [0, 1).$$

Let AP be the sorted list of allowed PCIs, with $m = |AP|$.

Deterministic decoding (initial assignment):

$$x_v \;=\; AP[\lfloor k_v \cdot m \rfloor]\,.$$

- This decoding is fast, but it can yield **infeasible** assignments.
- Feasibility is enforced by an embedded repair + local improvement phase.

Non-dominated Sorting Biased Random-Key Genetic Algorithm

Decoder step 1: fast feasibility repair

We quantify hard-constraint violations via three counters:

$$v_{\text{col}}(\mathbf{x}) = \sum_{(u,v)\in E} \mathbb{I}(x_u = x_v),$$

$$v_{\text{tn}}(\mathbf{x}) = \sum_{(u,v)\in T} \mathbb{I}\big(x_u \equiv x_v \pmod{k}\big),$$

$$v_{\text{conf}}(\mathbf{x}) = \sum_{(u,v)\in S} \mathbb{I}(x_u = x_v).$$

A solution is feasible iff $v_{\text{col}} = v_{\text{tn}} = v_{\text{conf}} = 0$.

Greedy best-effort repair:

- Build the set of *violated* cells $U \subseteq V$.
- Pick $v \in U$ with the most incident violations.
- Test all $p \in AP$ and choose $x_v \leftarrow p$ that minimizes $v_{\text{col}} + v_{\text{tn}} + v_{\text{conf}}$.
- Repeat while improving the total violation count.

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Decoder step 1b: penalize residual infeasibility

If repair cannot remove all violations, we penalize the first objective:

$$\tilde{f}_1(\mathbf{x}) = f_1(\mathbf{x}) + M \cdot \big(v_{\text{col}}(\mathbf{x}) + v_{\text{tn}}(\mathbf{x}) + v_{\text{conf}}(\mathbf{x})\big),$$

with a large constant M (set as $M = |N| + 1$ in the paper).

Returned fitness is:

$$\big(\tilde{f}_1(\mathbf{x}),\ f_2(\mathbf{x}),\ f_3(\mathbf{x})\big).$$

- Any remaining hard-constraint violation makes $\tilde{f}_1$ worse than the worst feasible f_1 value.
- This prevents infeasible solutions from dominating feasible ones.

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Decoder step 2: lexicographically guided local search

Once feasible, we run local search over **feasible single-cell moves**:

$$x_v \leftarrow p, \quad p \in AP \setminus \{x_v\} \text{ and move preserves feasibility.}$$

Move acceptance (inside the decoder):
first-improvement under a lexicographic order

$$(a_1, a_2, a_3) \prec_{\text{lex}} (b_1, b_2, b_3) \;\Leftrightarrow\; \begin{cases} a_1 < b_1, \text{ or} \\ a_1 = b_1 \;\wedge\; a_2 < b_2, \text{ or} \\ a_1 = b_1 \;\wedge\; a_2 = b_2 \;\wedge\; a_3 < b_3. \end{cases}$$

- Local search prioritizes operational objectives in the order (f_1, f_2, f_3).
- Example (from the paper): increasing f_3 is accepted only if it improves f_2 or f_1.

Non-dominated Sorting Biased Random-Key Genetic Algorithm

Decoder step 3: genotype feedback

After local search improves **x**, we **re-encode** the solution back into the chromosome.

If $x_v = AP[j]$ for $j \in \{0, \ldots, |AP| - 1\}$, set:

$$k_v \leftarrow \frac{j + 0.5}{|AP|} \quad \Rightarrow \quad \lfloor k_v \cdot |AP| \rfloor = j.$$

- The decoder becomes a **memetic operator**: improvements are not lost.
- Evolutionary operators can now recombine and propagate locally improved building blocks.

Non-dominated Sorting Biased Random-Key Genetic Algorithm

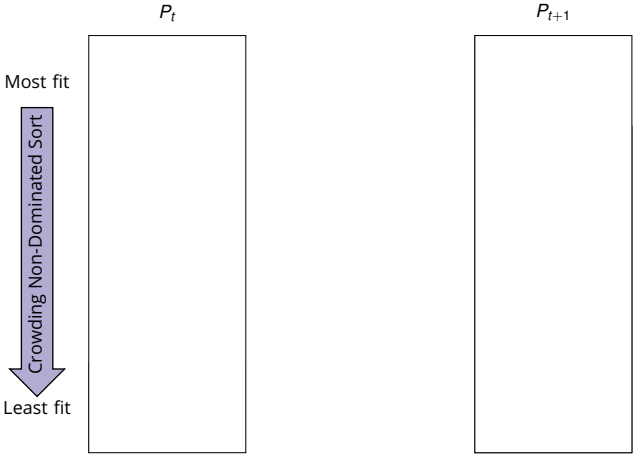
Evolution

P_t

P_{t+1}

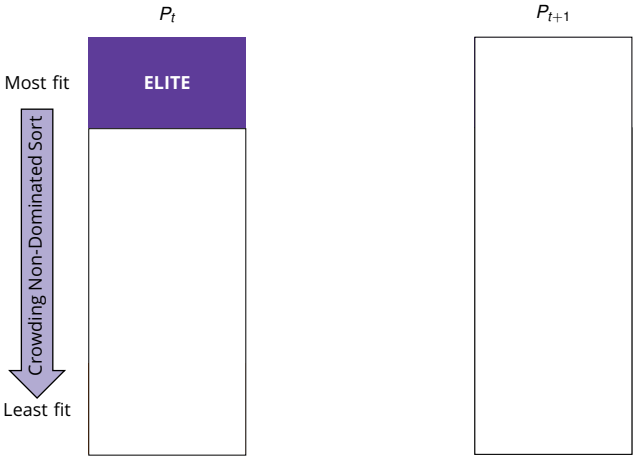
Non-dominated Sorting Biased Random-Key Genetic Algorithm

Evolution



Non-dominated Sorting Biased Random-Key Genetic Algorithm

Evolution



Non-dominated Sorting Biased Random-Key Genetic Algorithm

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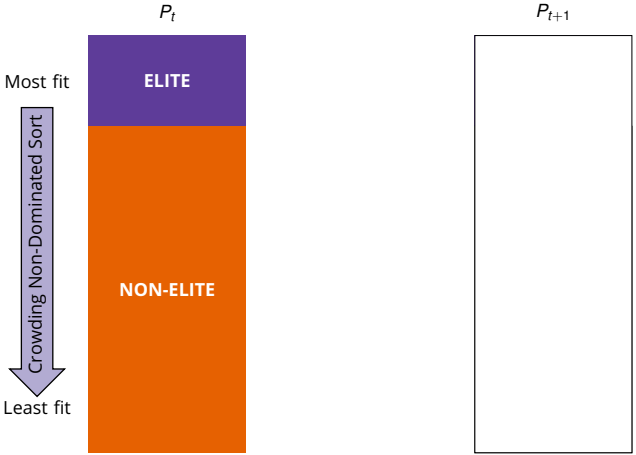
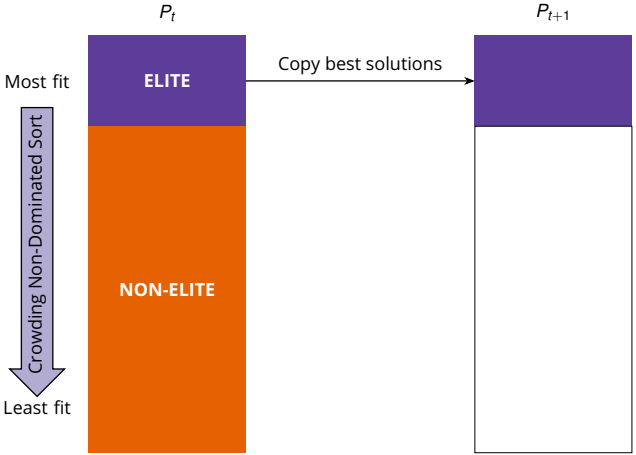


Non-dominated Sorting Biased Random-Key Genetic Algorithm

Evolution



Non-dominated Sorting Biased Random-Key Genetic Algorithm

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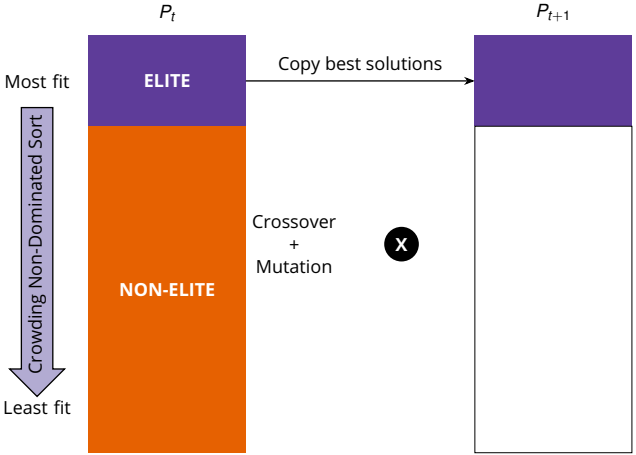

Non-dominated Sorting Biased Random-Key Genetic Algorithm

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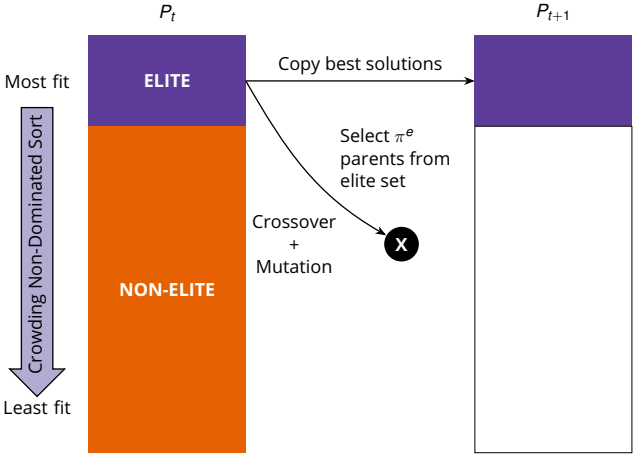

Non-dominated Sorting Biased Random-Key Genetic Algorithm

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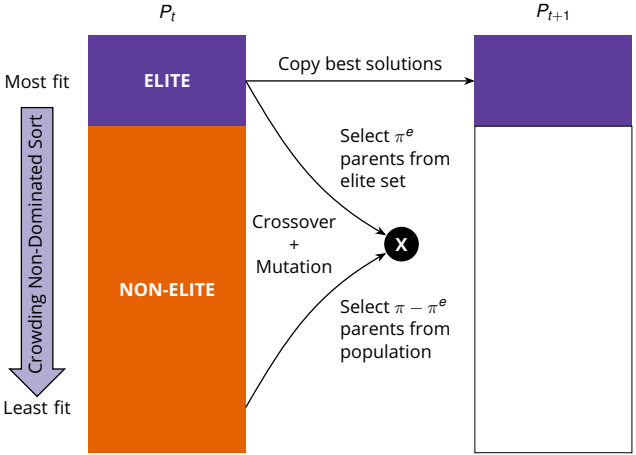

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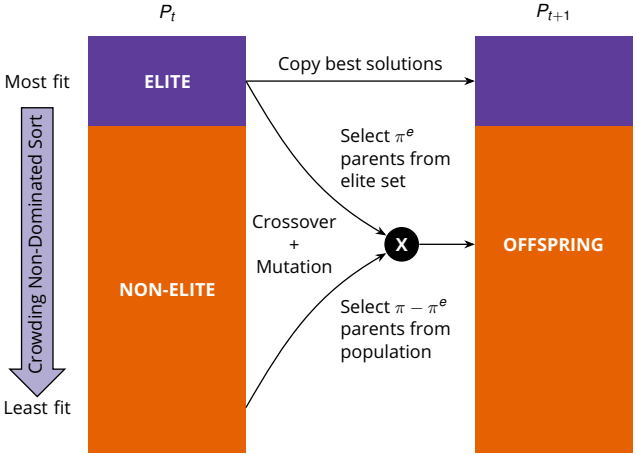
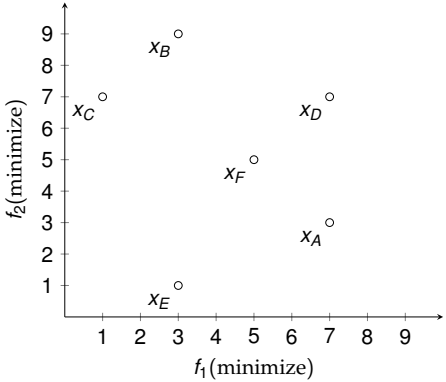


Non-dominated Sorting Biased Random-Key Genetic Algorithm

Non-dominated Sort



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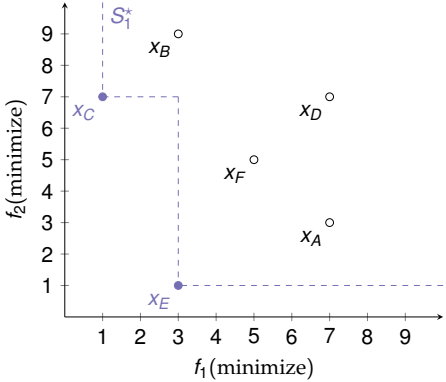
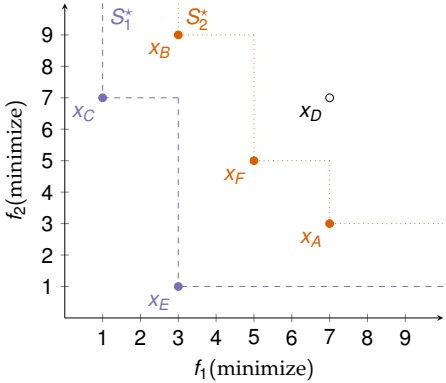


$$S_1^\star = \{x_C, x_E\}$$

Non-dominated Sorting Biased Random-Key Genetic Algorithm

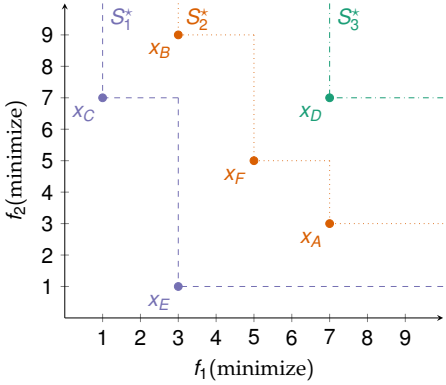
Non-dominated Sort



$S_1^\star = \{x_C, x_E\}$ $\quad$ $S_2^\star = \{x_A, x_B, x_F\}$

Non-dominated Sorting Biased Random-Key Genetic Algorithm

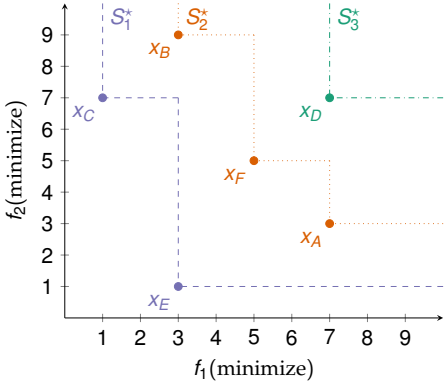
Non-dominated Sort



$S_1^\star = \{x_C, x_E\}$ $S_2^\star = \{x_A, x_B, x_F\}$ $S_3^\star = \{x_D\}$

Non-dominated Sorting Biased Random-Key Genetic Algorithm

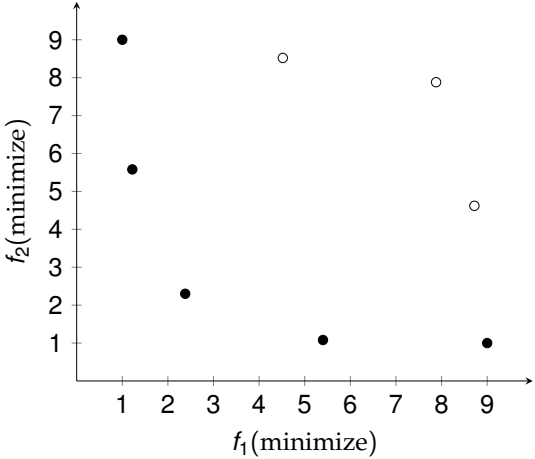
Non-dominated Sort



$$S_1^\star = \{x_C, x_E\} \prec S_2^\star = \{x_A, x_B, x_F\} \prec S_3^\star = \{x_D\}$$

Non-dominated Sorting Biased Random-Key Genetic Algorithm

Crowding Distance



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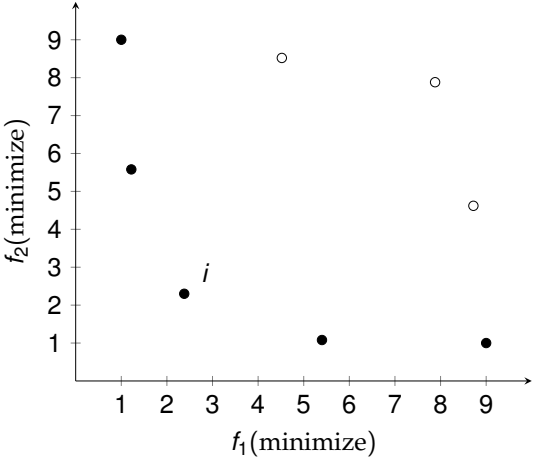
Crowding Distance



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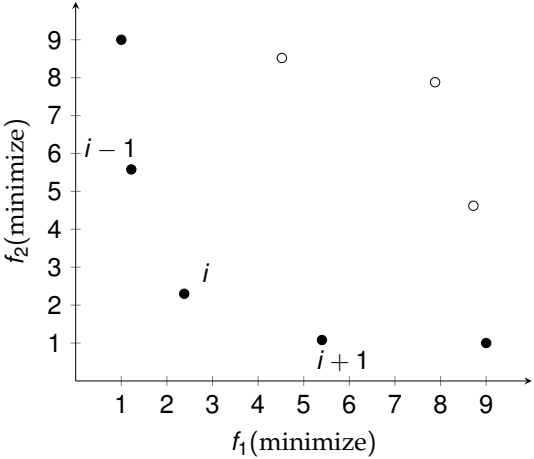
Crowding Distance



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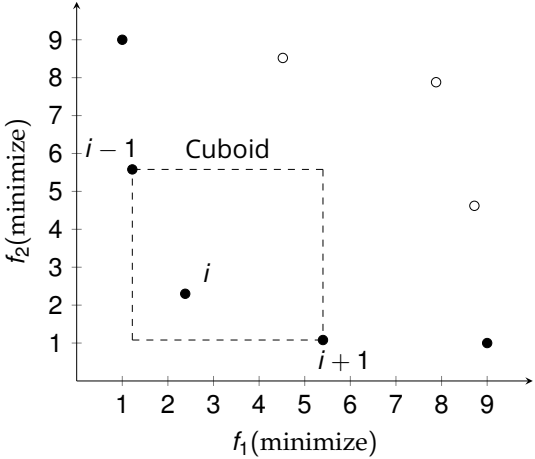
Crowding Distance



Non-dominated Sorting Biased Random-Key Genetic Algorithm

Multi-Parent Biased Crossover

Non-dominated Sorting Biased Random-Key Genetic Algorithm

Multi-Parent Biased Crossover

Parent 1	0.16	0.53	0.85	0.76	0.99	0.06	0.38	# 03
Parent 2	0.12	0.63	0.21	0.69	0.08	0.01	0.90	# 05

Non-dominated Sorting Biased Random-Key Genetic Algorithm

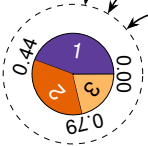
Multi-Parent Biased Crossover

Parent 1	0.16	0.53	0.85	0.76	0.99	0.06	0.38	# 03
Parent 2	0.12	0.63	0.21	0.69	0.08	0.01	0.90	# 05
Parent 3	0.91	0.68	0.52	0.02	0.18	0.34	0.22	# 13

Non-dominated Sorting Biased Random-Key Genetic Algorithm

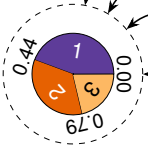
Multi-Parent Biased Crossover

Parent 1	0.16	0.53	0.85	0.76	0.99	0.06	0.38	# 03
Parent 2	0.12	0.63	0.21	0.69	0.08	0.01	0.90	# 05
Parent 3	0.91	0.68	0.52	0.02	0.18	0.34	0.22	# 13



Non-dominated Sorting Biased Random-Key Genetic Algorithm

Multi-Parent Biased Crossover



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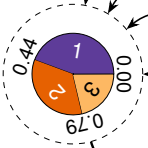
Random number

Non-dominated Sorting Biased Random-Key Genetic Algorithm

Multi-Parent Biased Crossover



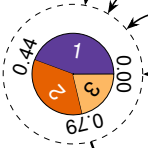
Parent 1	0.16	0.53	0.85	0.76	0.99	0.06	0.38	# 03
Parent 2	0.12	0.63	0.21	0.69	0.08	0.01	0.90	# 05
Parent 3	0.91	0.68	0.52	0.02	0.18	0.34	0.22	# 13

Random number

Offspring

Non-dominated Sorting Biased Random-Key Genetic Algorithm

Multi-Parent Biased Crossover



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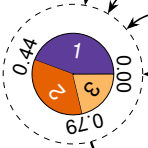
Parent 2	0.12	0.63	0.21	0.69	0.08	0.01	0.90	# 05
Parent 3	0.91	0.68	0.52	0.02	0.18	0.34	0.22	# 13
Random number	0.21							
Offspring	0.16							

Non-dominated Sorting Biased Random-Key Genetic Algorithm

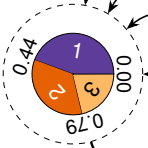
Multi-Parent Biased Crossover



Parent 1	0.16	0.53	0.85	0.76	0.99	0.06	0.38	# 03
Parent 2	0.12	0.63	0.21	0.69	0.08	0.01	0.90	# 05
Parent 3	0.91	0.68	0.52	0.02	0.18	0.34	0.22	# 13
Random number	0.21	0.94						
Offspring	0.16	0.68						

Non-dominated Sorting Biased Random-Key Genetic Algorithm

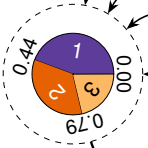
Multi-Parent Biased Crossover



Parent 1	0.16	0.53	0.85	0.76	0.99	0.06	0.38	# 03
Parent 2	0.12	0.63	0.21	0.69	0.08	0.01	0.90	# 05
Parent 3	0.91	0.68	0.52	0.02	0.18	0.34	0.22	# 13
Random number	0.21	0.94	0.52					
Offspring	0.16	0.68	0.21					

Non-dominated Sorting Biased Random-Key Genetic Algorithm

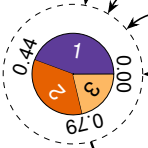
Multi-Parent Biased Crossover



Parent 1	0.16	0.53	0.85	0.76	0.99	0.06	0.38	# 03
Parent 2	0.12	0.63	0.21	0.69	0.08	0.01	0.90	# 05
Parent 3	0.91	0.68	0.52	0.02	0.18	0.34	0.22	# 13
Random number	0.21	0.94	0.52	0.02				
Offspring	0.16	0.68	0.21	0.76				

Non-dominated Sorting Biased Random-Key Genetic Algorithm

Multi-Parent Biased Crossover



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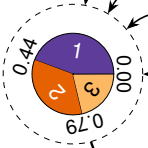
Parent 2	0.12	0.63	0.21	0.69	0.08	0.01	0.90	# 05
Parent 3	0.91	0.68	0.52	0.02	0.18	0.34	0.22	# 13
Random number	0.21	0.94	0.52	0.02	0.75			
Offspring	0.16	0.68	0.21	0.76	0.08			

Non-dominated Sorting Biased Random-Key Genetic Algorithm

Multi-Parent Biased Crossover



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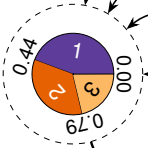
Parent 2	0.12	0.63	0.21	0.69	0.08	0.01	0.90	# 05
Parent 3	0.91	0.68	0.52	0.02	0.18	0.34	0.22	# 13
Random number	0.21	0.94	0.52	0.02	0.75	0.17		
Offspring	0.16	0.68	0.21	0.76	0.08	0.06		

Non-dominated Sorting Biased Random-Key Genetic Algorithm

Multi-Parent Biased Crossover



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Parent 3	0.91	0.68	0.52	0.02	0.18	0.34	0.22	# 13
Random number	0.21	0.94	0.52	0.02	0.75	0.17	0.14	
Offspring	0.16	0.68	0.21	0.76	0.08	0.06	0.38	

Non-dominated Sorting Biased Random-Key Genetic Algorithm

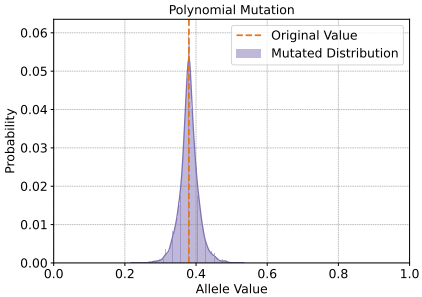
Polynomial Mutation

Applied with probability p_m to induce an expected perturbation of $\mathcal{O}(\frac{1}{\eta_m})$ in a gene.

Generates a new allele as follows:

$$k'_v = k_v + \delta,$$

where δ is a perturbation value calculated from a random number $u \in [0, 1)$, as follows:



$$\delta = \begin{cases} (2u + (1-2u)(1-k_v)^{\eta_m+1})^{\frac{1}{\eta_m+1}} - 1 & \text{if } u < \frac{1}{2}, \\ 1 - (2(1-u) + 2(u-\frac{1}{2})(k_v)^{\eta_m+1})^{\frac{1}{\eta_m+1}} & \text{otherwise.} \end{cases}$$

$$k_v = 0.38 \Rightarrow k'_v \approx 0.38 \pm 0.03$$
$$\eta_m = 50.00 \Rightarrow |k'_v - k_v| \approx 0.02 \pm 0.02$$

Non-dominated Sorting Biased Random-Key Genetic Algorithm

Other features

- Genotypic diversity.
- Migration.
- Implicit Path-Relinking.
- Shaking.
- Partial reset.

Computational Experiments

Experimental Setup

We performed 5 independent runs of each algorithm on each test problem on a computer with a CPU of 3.30GHz and 32GB of RAM with a time limit of 15 minutes.

Experimental Setup

NS-BRKGA parameter settings

Parameter	**Value**
Population size $\lvert P \rvert$ (per subpopulation)	300
Parent count π (elitist π_e)	3 (with $\pi_e = 2$ elites)
Elite fraction p^e_{min} – p^e_{max}	0.10 – 0.30
Bias function $\Phi(r)$	$1/\sqrt{r}$
Mutation probability p_m (index η_m)	0.01 ($\eta_m = 50$)
Number of subpopulations p	3
Migration interval (gen), size k	200 gen, $k = 30$ solutions
IPR interval (gen)	500 gen
Shaking trigger (stagnant gen)	200 gen ($p_{\text{shake}} = 0.33$, $\eta_{\text{shake}} = 20$)
Reset trigger (stagnant gen)	500 gen ($p_{\text{reset}} = 0.20$)

Benchmark Instances

- 291 real-world MOPCI instances (sizes from 30 to 4125 cells).
- Instance types (PCI groups):
  - `full`: $AP = \{0, \ldots, 1007\}$ (least constrained),
  - `orig`: $AP = \{\min(\mathbf{O}), \ldots, \max(\mathbf{O})\}$,
  - `squz`: minimal contiguous AP that still admits feasibility (most constrained).

Baseline Algorithms

The proposed NS-BRKGA is benchmarked against the following algorithms[1]:

- **NSGA-II**: The Elitist Non-dominated Sorting Genetic Algorithm.
- **NSPSO**: The Non-dominated Sorting Particle Swarm Optimizer.
- **MOEA/D-DE**: The Multi-Objective Evolutionary Algorithm by Decomposition with Differential Evolution.
- **MHACO**: The Multi-objective Hypervolume-based Ant Colony Optimizer.
- **IHS**: The Improved Harmony Search.

[1] Implementations at: https://esa.github.io/pagmo2/

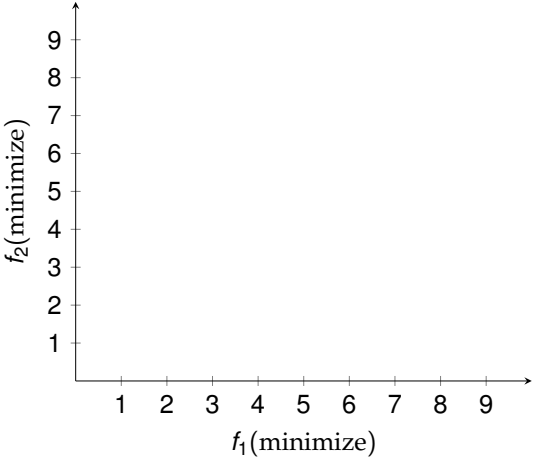
Performance Indicators

Two complementary quality indicators were employed:

1. **Normalized Modified Inverted Generational Distance** ($NIGD^{+}$),
2. **Hypervolume Ratio** (HVR).

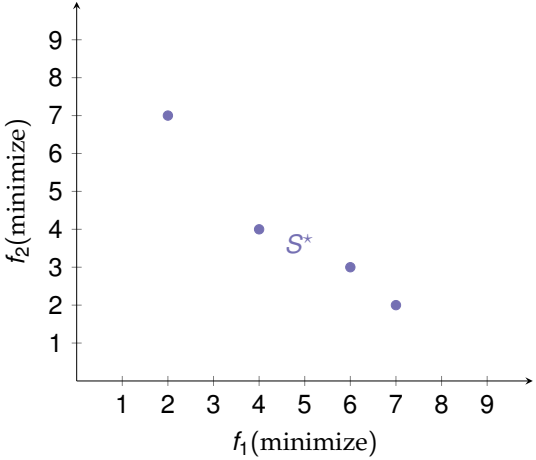
Performance Indicators

Normalized Modified Inverted Generational Distance ($NIGD^+$)



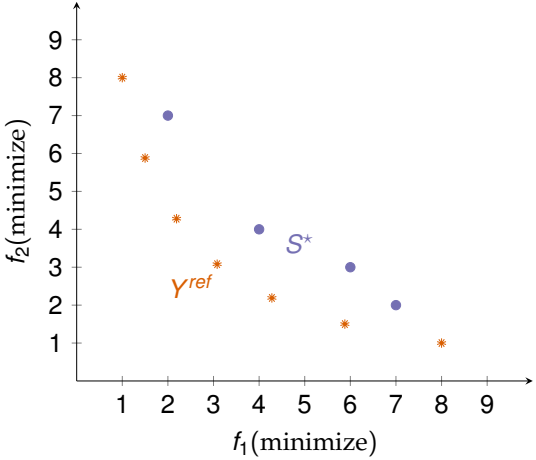
Performance Indicators

Normalized Modified Inverted Generational Distance ($NIGD^+$)



Performance Indicators

Normalized Modified Inverted Generational Distance ($NIGD^+$)



Performance Indicators

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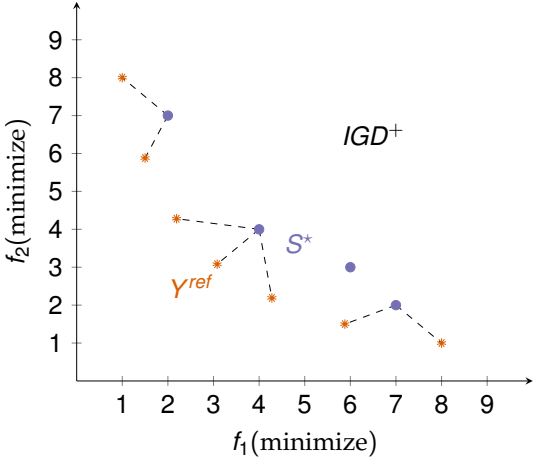

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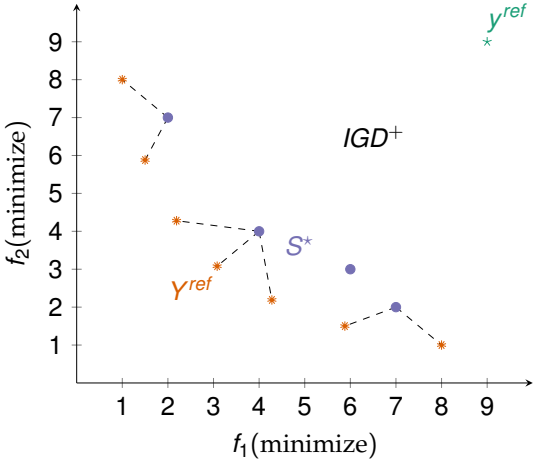

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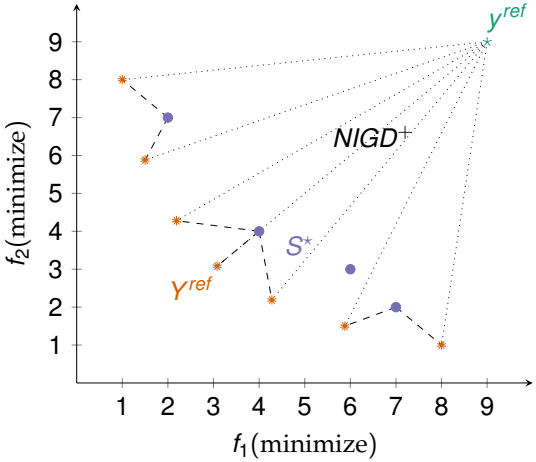
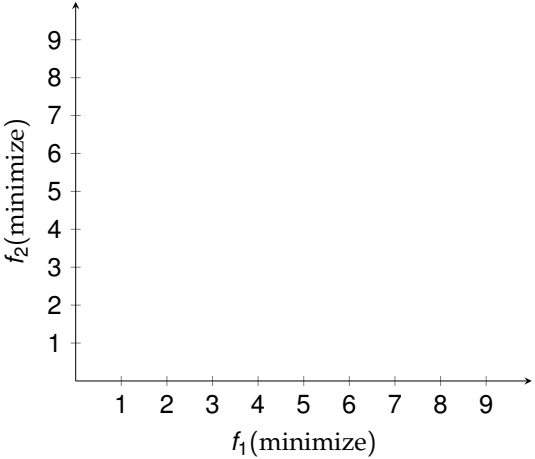


Performance Indicators

Hypervolume Ratio (*HVR*)



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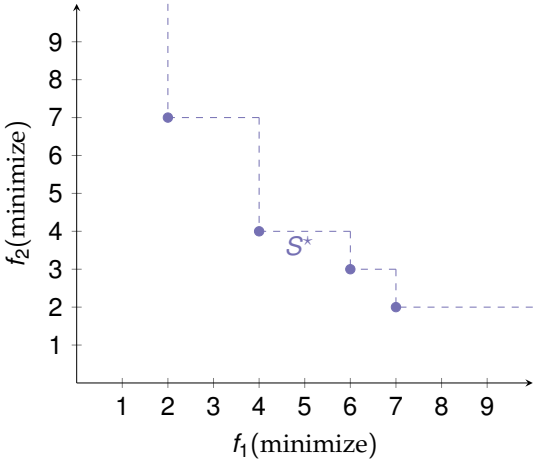

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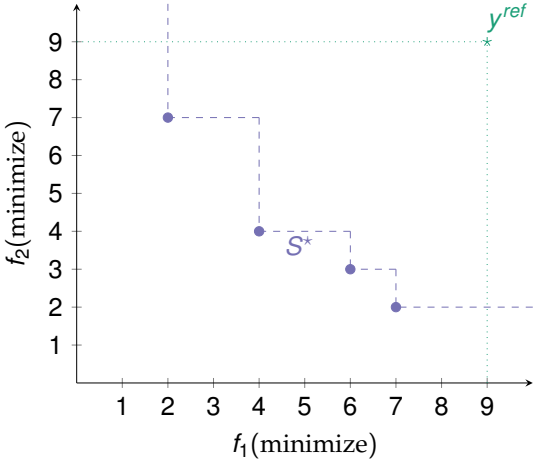

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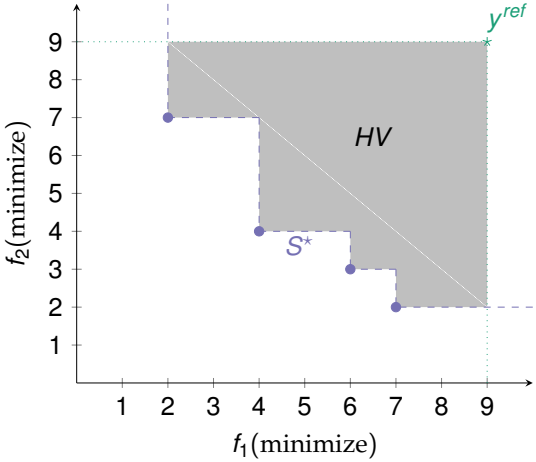

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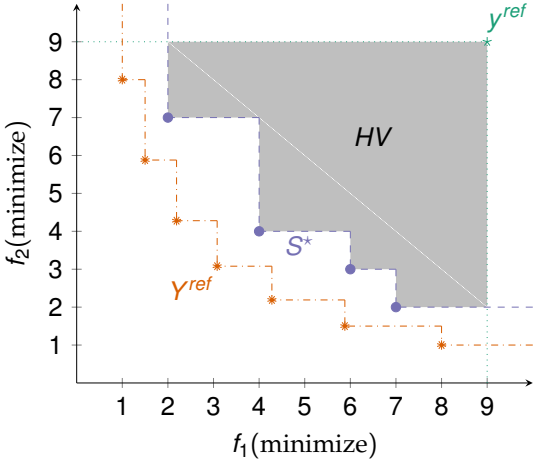
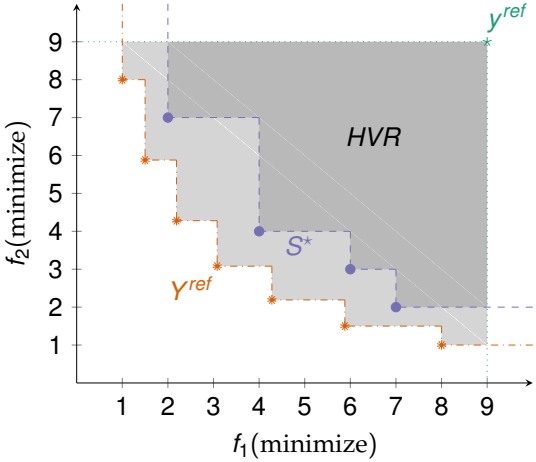


Performance Indicators

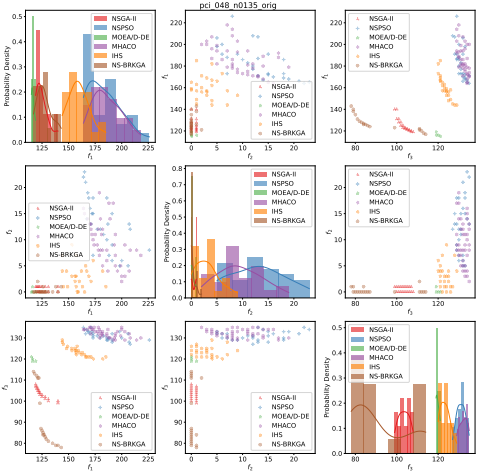
Hypervolume Ratio (*HVR*)



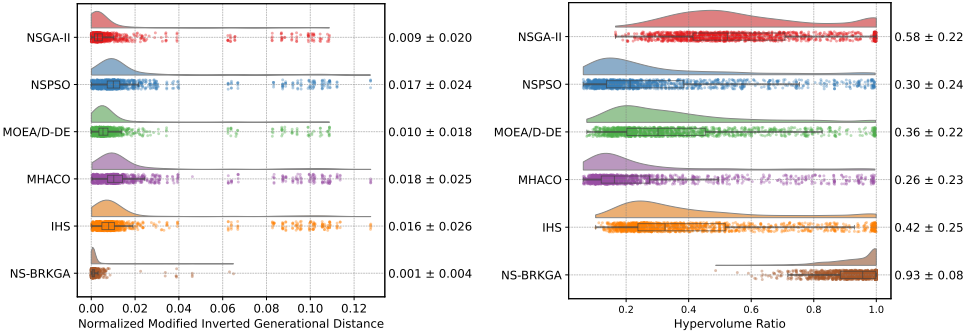
Results
&
Analysis

Results for a single instance

pci_048_n0135_orig_best

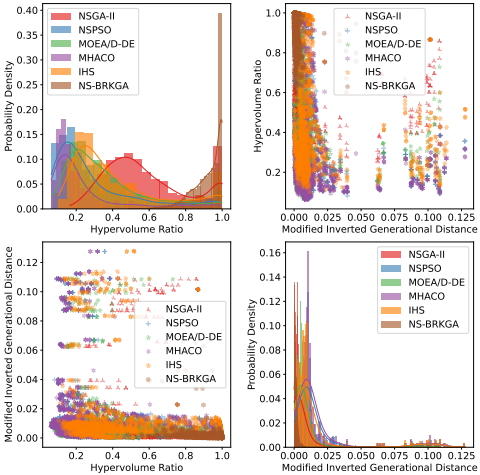


Overall Performance Comparison

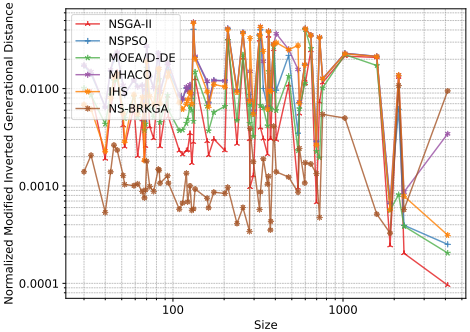


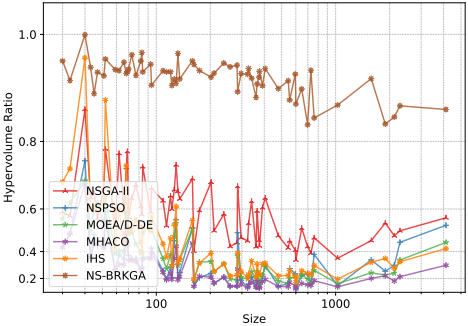
- NS-BRKGA attains the best mean ranks on **both** $NIGD^+$ and HVR.
- Reported means: $NIGD^+ \approx 0.001$ and $HVR \approx 0.93$ (NSGA-II: 0.009 and 0.58).

Overall Performance Comparison



Scalability with Problem Size





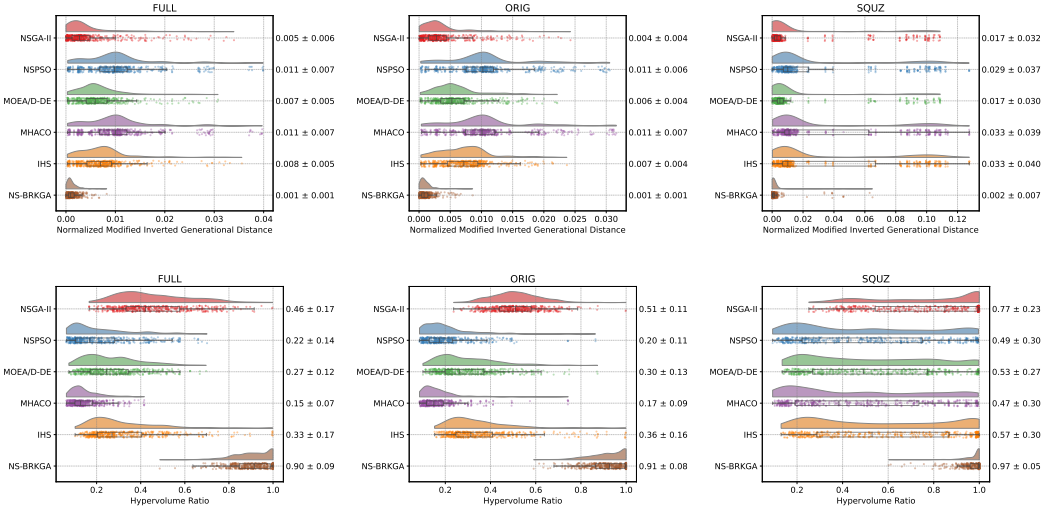
- As $|V|$ grows, baseline methods degrade, while NS-BRKGA remains stable.
- On the largest instances ($|V| = 4125$), NS-BRKGA maintains mean $HVR \approx 0.87 \pm 0.08$ (NSGA-II $\approx 0.6 \pm 0.1$).

Robustness Across Instance Types

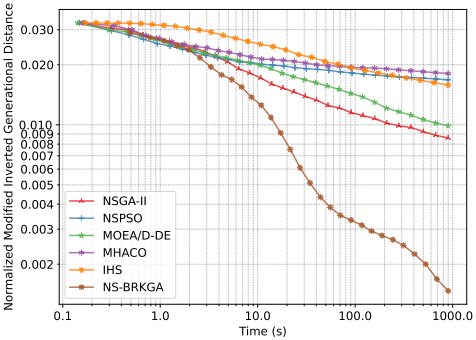
Algorithm	$NIGD^+$ (↓)			HVR (↑)		
	`full`	`orig`	`squz`	`full`	`orig`	`squz`
NSGA-II	0.005 ± 0.006	0.004 ± 0.004	0.017 ± 0.032	0.46 ± 0.17	0.51 ± 0.11	0.77 ± 0.23
NSPSO	0.011 ± 0.007	0.011 ± 0.006	0.029 ± 0.037	0.22 ± 0.14	0.20 ± 0.11	0.49 ± 0.30
MOEA/D-DE	0.007 ± 0.005	0.006 ± 0.004	0.017 ± 0.030	0.27 ± 0.12	0.30 ± 0.13	0.53 ± 0.27
MHACO	0.011 ± 0.007	0.011 ± 0.007	0.033 ± 0.039	0.15 ± 0.07	0.17 ± 0.09	0.47 ± 0.30
IHS	0.008 ± 0.005	0.007 ± 0.004	0.033 ± 0.040	0.33 ± 0.17	0.36 ± 0.16	0.57 ± 0.30
NS-BRKGA	$\mathbf{0.001 \pm 0.001}$	$\mathbf{0.001 \pm 0.001}$	$\mathbf{0.002 \pm 0.007}$	$\mathbf{0.90 \pm 0.09}$	$\mathbf{0.91 \pm 0.08}$	$\mathbf{0.97 \pm 0.05}$

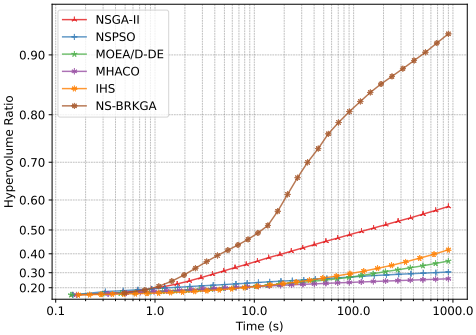
- NS-BRKGA is the top performer on all instance types.
- On `squz`: mean $HVR \approx 0.97$ and mean $NIGD^+ \approx 0.002$.

Robustness Across Instance Types



Convergence & Runtime Performance





- NS-BRKGA reaches high-quality fronts **very early** (within $\sim$ 30s: $HVR \approx 0.7 \pm 0.2$).
- Baselines often do not match this quality even after the full 15-minute budget.

Conclusion

Conclusion

- **MOPCI** is a practical LTE/5G optimization problem involving both **technological constraints** and multi-objective **operational concerns**.
- **Methodological contribution:** NS-BRKGA integrates a random-key evolutionary framework with a problem-specific decoder featuring **feasibility repair**, **lexicographically guided local search**, and **genotype feedback**.
- **Experimental evidence:** Across 291 real-world instances and five representative MOEAs, NS-BRKGA consistently achieves **strong Pareto-front quality** ($HVR \approx 0.93$, $NIGD^{+} \approx 0.001$), **scalability**, **robustness**, and **fast convergence** ($\sim$ 30s).
- **Take-home message:** NS-BRKGA provides network operators with a highly practical way to quickly generate high-quality PCI plans that balance **radio performance**, **rollout safety**, and **operational stability**.

That’s all, folks!